

Data

speed of light in free space,

$$c = 3.00 \times 10^8 \text{ m s}^{-1}$$

permeability of free space,

$$\mu_0 = 4\pi \times 10^{-7} \text{ H m}^{-1}$$

permittivity of free space,

$$\epsilon_0 = 8.85 \times 10^{-12} \text{ F m}^{-1}$$

elementary charge,

$$e = 1.60 \times 10^{-19} \text{ C}$$

the Planck constant,

$$h = 6.63 \times 10^{-34} \text{ J s}$$

unified atomic mass constant,

$$u = 1.66 \times 10^{-27} \text{ kg}$$

rest mass of electron,

$$m_e = 9.11 \times 10^{-31} \text{ kg}$$

rest mass of proton,

$$m_p = 1.67 \times 10^{-27} \text{ kg}$$

molar gas constant,

$$R = 8.31 \text{ J K}^{-1} \text{ mol}^{-1}$$

the Avogadro constant,

$$N_A = 6.02 \times 10^{23} \text{ mol}^{-1}$$

the Boltzmann constant,

$$k = 1.38 \times 10^{-23} \text{ J K}^{-1}$$

gravitational constant,

$$G = 6.67 \times 10^{-11} \text{ N m}^2 \text{ kg}^{-2}$$

acceleration of free fall,

$$g = 9.81 \text{ m s}^{-2}$$

Formulae

uniformly accelerated motion,

$$s = ut + \frac{1}{2}at^2$$

$$v^2 = u^2 + 2as$$

work done on/by a gas,

$$W = p\Delta V$$

gravitational potential,

$$\phi = -\frac{Gm}{r}$$

simple harmonic motion,

$$a = -\omega^2 x$$

velocity of particle in s.h.m.,

$$v = v_0 \cos \omega t$$

$$v = \pm \omega \sqrt{(x_0^2 - x^2)}$$

resistors in series,

$$R = R_1 + R_2 + \dots$$

resistors in parallel,

$$1/R = 1/R_1 + 1/R_2 + \dots$$

electric potential,

$$V = \frac{Q}{4\pi\epsilon_0 r}$$

capacitors in series,

$$1/C = 1/C_1 + 1/C_2 + \dots$$

capacitors in parallel,

$$C = C_1 + C_2 + \dots$$

energy of charged capacitor,

$$W = \frac{1}{2} QV$$

alternating current/voltage,

$$x = x_0 \sin \omega t$$

hydrostatic pressure,

$$p = \rho gh$$

pressure of an ideal gas,

$$p = \frac{1}{3} \frac{Nm}{V} \langle c^2 \rangle$$

radioactive decay,

$$x = x_0 \exp(-\lambda t)$$

decay constant,

$$\lambda = \frac{0.693}{t_{1/2}}$$

critical density of matter in the Universe,

$$\rho_0 = \frac{3H_0^2}{8\pi G}$$

equation of continuity,

$$Av = \text{constant}$$

Bernoulli equation (simplified),

$$p_1 + \frac{1}{2}\rho v_1^2 = p_2 + \frac{1}{2}\rho v_2^2$$

Stokes' law,

$$F = Ar\eta v$$

Reynolds' number,

$$R_e = \frac{\rho v r}{\eta}$$

drag force in turbulent flow,

$$F = Br^2 \rho v^2$$

Answer **all** the questions in the spaces provided.

For
Examiner's
Use

- 1 (a) (i) Define *pressure*.

.....
..... [1]

- (ii) State the units of pressure in base units.

..... [1]

- (b) The pressure p at a depth h in an incompressible fluid of density ρ is given by

$$p = \rho gh,$$

where g is the acceleration of free fall.

Use base units to check the homogeneity of this equation.

.....
.....
.....
..... [3]

- 2 (a) Explain what is meant by the *centre of gravity* of a body.

.....

.....

..... [2]

For
Examiner's
Use

- (b) An irregularly-shaped piece of cardboard is hung freely from one point near its edge, as shown in Fig. 2.1.

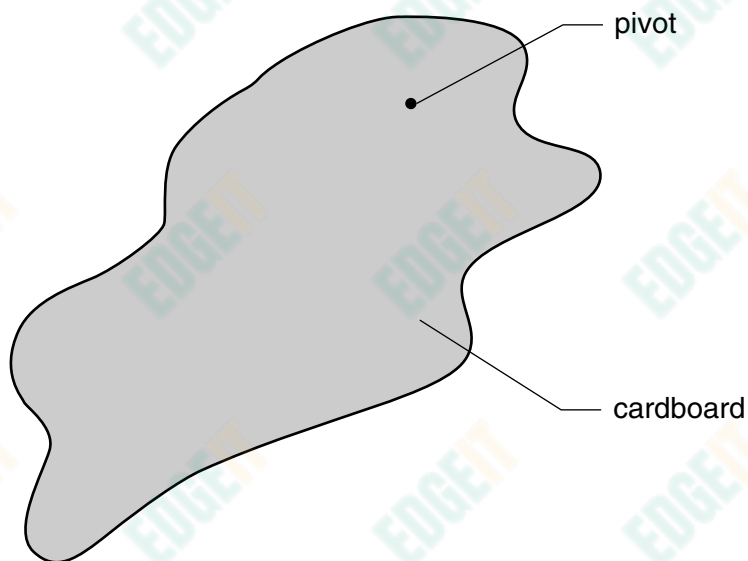


Fig. 2.1

Explain why the cardboard will come to rest with its centre of gravity vertically below the pivot. You may draw on Fig. 2.1 if you wish.

.....

.....

..... [2]

- 3 A stone on a string is made to travel along a horizontal circular path, as shown in Fig. 3.1.

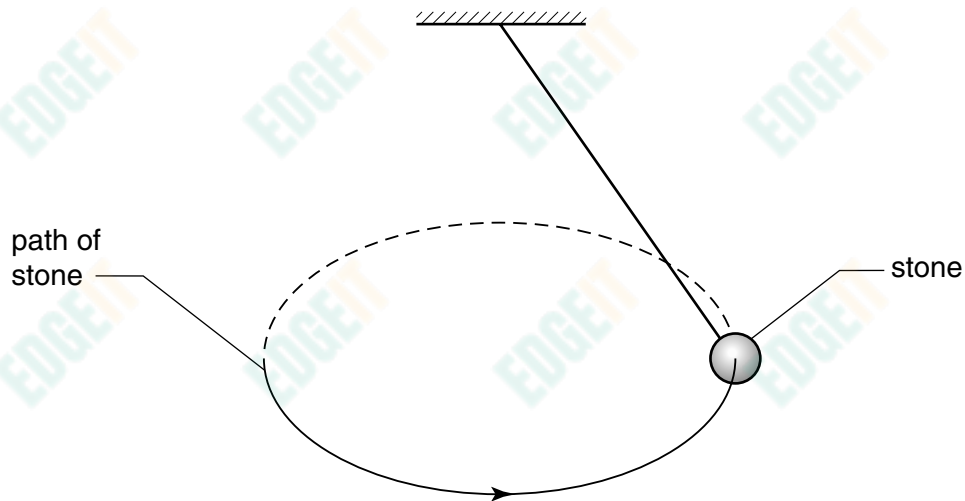


Fig. 3.1

The stone has a constant speed.

- (a) Define *acceleration*.

.....
 [1]

- (b) Use your definition to explain whether the stone is accelerating.

.....

 [2]

- (c) The stone has a weight of 5.0 N. When the string makes an angle of 35° to the vertical, the tension in the string is 6.1 N, as illustrated in Fig. 3.2.

For
Examiner's
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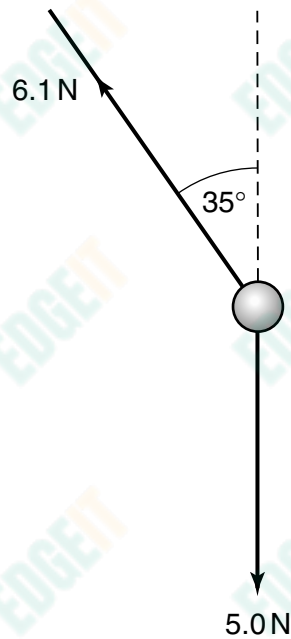


Fig. 3.2

Determine the resultant force acting on the stone in the position shown.

magnitude of force = N

direction of force..... [4]

- 4 A trolley of mass 930 g is held on a horizontal surface by means of two springs, as shown in Fig. 4.1.

For
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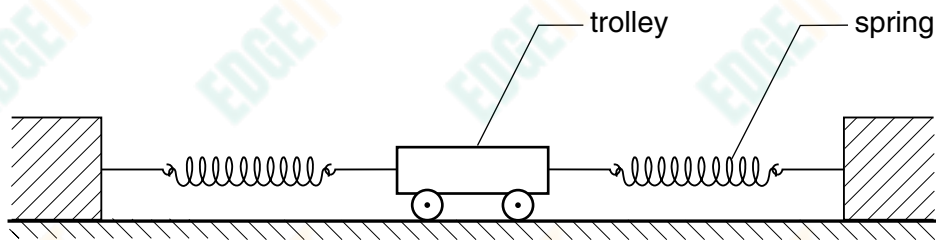


Fig. 4.1

The variation with time t of the speed v of the trolley for the first 0.60 s of its motion is shown in Fig. 4.2.

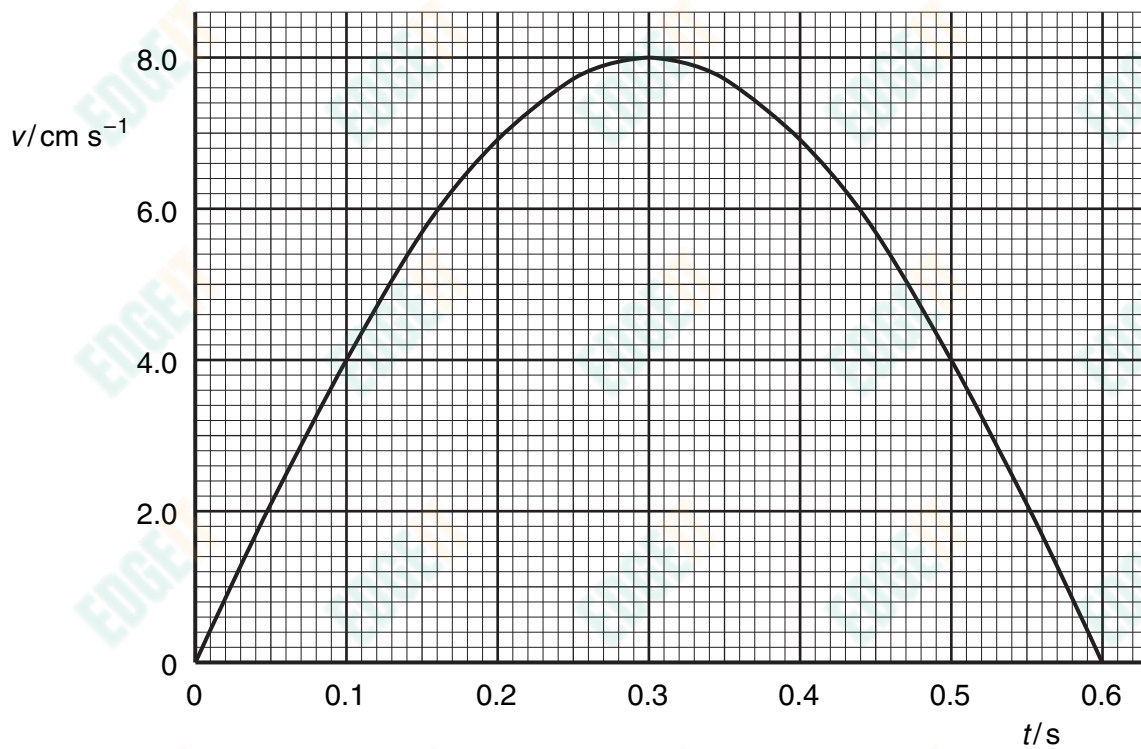


Fig. 4.2

- (a) Use Fig. 4.2 to determine
(i) the initial acceleration of the trolley,

acceleration = m s^{-2} [2]

- (ii) the distance moved during the first 0.60 s of its motion.

For
Examiner's
Use

distance = m [3]

- (b) (i) Use your answer to (a)(i) to determine the resultant force acting on the trolley at time $t = 0$.

force = N [2]

- (ii) Describe qualitatively the variation with time of the resultant force acting on the trolley during the first 0.60 s of its motion.

.....
.....
.....
..... [3]

- 5 Fig. 5.1 shows the variation with time t of the displacements x_A and x_B at a point P of two sound waves A and B.

For
Examiner's
Use

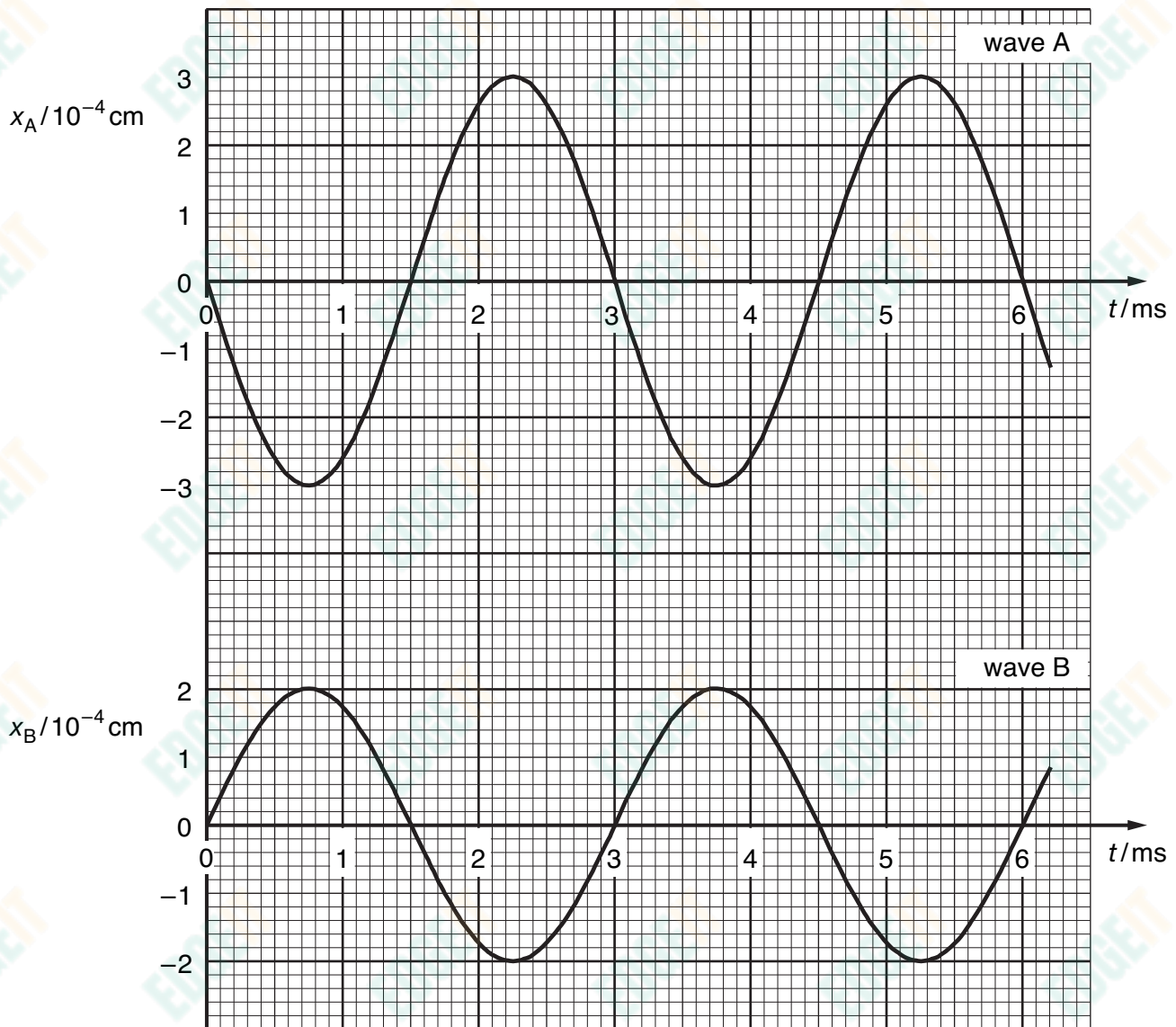


Fig. 5.1

- (a) By reference to Fig. 5.1, state one similarity and one difference between these two waves.

similarity:

difference: [2]

- (b) State, with a reason, whether the two waves are coherent.

.....

..... [1]

(c) The intensity of wave A alone at point P is I .

(i) Show that the intensity of wave B alone at point P is $\frac{4}{9}I$.

[2]

(ii) Calculate the resultant intensity, in terms of I , of the two waves at point P.

resultant intensity = I [2]

(d) Determine the resultant displacement for the two waves at point P

(i) at time $t = 3.0$ ms,

resultant displacement = cm [1]

(ii) at time $t = 4.0$ ms.

resultant displacement = cm [2]

- 6 Two horizontal metal plates X and Y are at a distance 0.75 cm apart. A positively charged particle of mass 9.6×10^{-15} kg is situated in a vacuum between the plates, as illustrated in Fig. 6.1.

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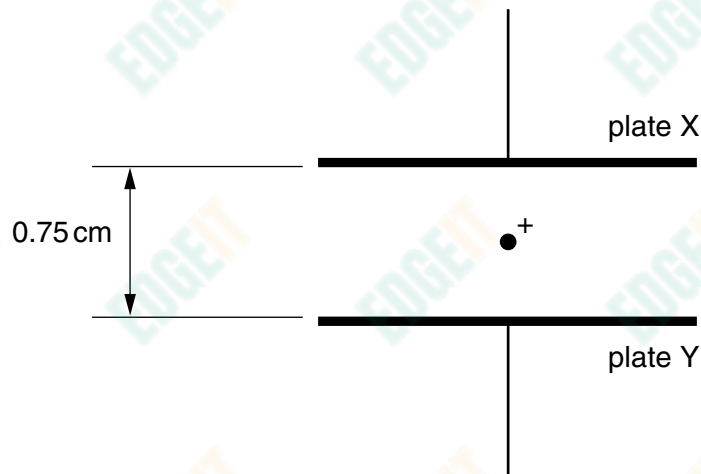


Fig. 6.1

The potential difference between the plates is adjusted until the particle remains stationary.

- (a) State, with a reason, which plate, X or Y, is positively charged.

.....

 [2]

- (b) The potential difference required for the particle to be stationary between the plates is found to be 630 V. Calculate

- (i) the electric field strength between the plates,

field strength = N C⁻¹ [2]

(ii) the charge on the particle.

For
Examiner's
Use

charge = C [3]

- 7 A battery of e.m.f. 4.50 V and negligible internal resistance is connected in series with a fixed resistor of resistance $1200\ \Omega$ and a thermistor, as shown in Fig. 7.1.

For
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Use

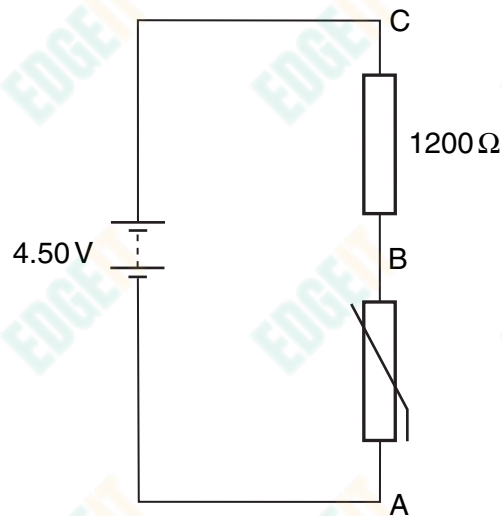


Fig. 7.1

- (a) At room temperature, the thermistor has a resistance of $1800\ \Omega$. Deduce that the potential difference across the thermistor (across AB) is 2.70 V.

[2]

- (b) A uniform resistance wire PQ of length 1.00 m is now connected in parallel with the resistor and the thermistor, as shown in Fig. 7.2.

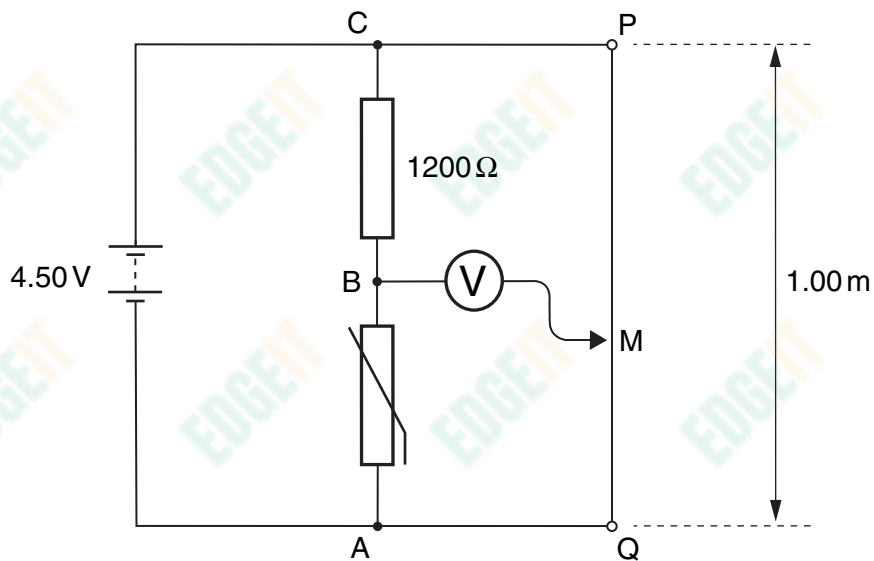


Fig. 7.2

A sensitive voltmeter is connected between point B and a moveable contact M on the wire.

- (i) Explain why, for constant current in the wire, the potential difference between any two points on the wire is proportional to the distance between the points.

.....

 [2]

- (ii) The contact M is moved along PQ until the voltmeter shows zero reading.

1. State the potential difference between the contact at M and the point Q.

potential difference = V [1]

2. Calculate the length of wire between M and Q.

length = cm [2]

- (iii) The thermistor is warmed slightly. State and explain the effect on the length of wire between M and Q for the voltmeter to remain at zero deflection.

.....

 [2]

- 8 (a) Explain the concept of *work*.

.....

.....

..... [2]

- (b) A table tennis ball falls vertically through air. Fig. 8.1 shows the variation of the kinetic energy E_K of the ball with distance h fallen. The ball reaches the ground after falling through a distance h_0 .

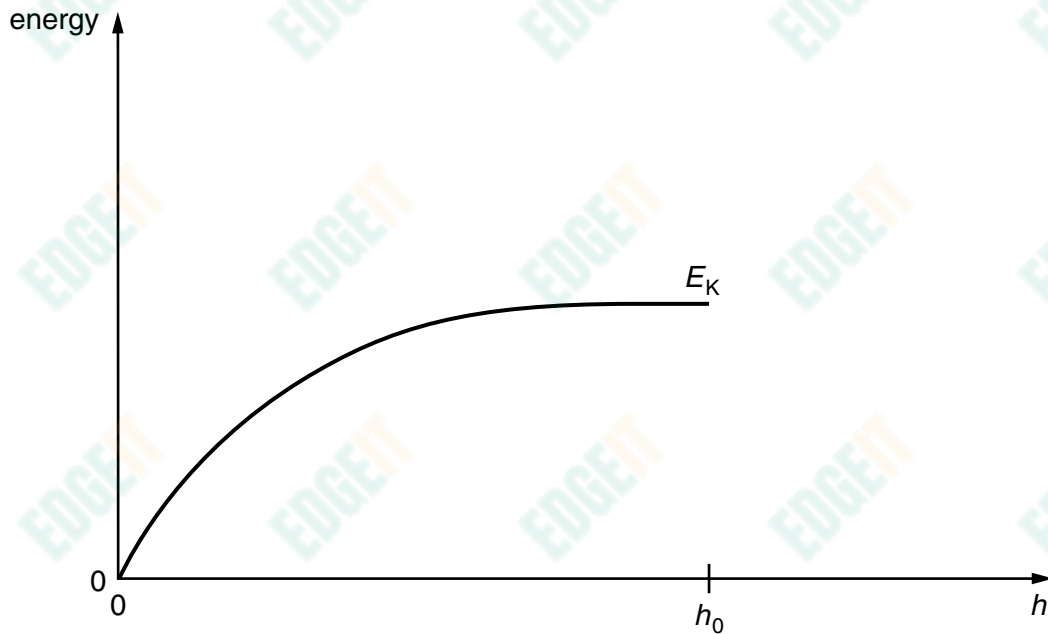


Fig. 8.1

- (i) Describe the motion of the ball.

.....

.....

.....

.....

..... [3]

- (ii) On Fig. 8.1, draw a line to show the variation with h of the gravitational potential energy E_P of the ball. At $h = h_0$, the potential energy is zero. [3]

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